

Métodos Computacionales

Dinámica Molecular

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Y COMPUTACIONAL**
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Horton Plains

Dinámica

Estudia el movimiento en relación con las causas que lo producen.

Ecuaciones de movimiento, por ejemplo. . . :

Newton

$$m \frac{d^2 \mathbf{r}}{dt^2} = -\nabla U(\mathbf{r})$$

Langevin $m \frac{\partial \mathbf{v}}{\partial t} = -\nabla U - \gamma \mathbf{v} + \sqrt{2\gamma k_B T} \boldsymbol{\eta}$

Browniano $\gamma \mathbf{v} = -\nabla U + \sqrt{2\gamma k_B T} \boldsymbol{\eta}$

Barostato de Andersen

$$\frac{d\mathbf{r}}{dt} = \frac{\mathbf{p}}{m} + \frac{\mathbf{r}}{3} \frac{d \ln V}{dt}$$

$$\frac{d\mathbf{p}}{dt} = -\nabla U - \frac{\mathbf{p}}{3} \frac{d \ln V}{dt}$$

$$\frac{d^2 V}{dt^2} = \frac{p_0}{M} + \frac{(\mathbf{p}^2/m - \nabla U)}{3MV}$$

DM acelerada por temperatura

$$\gamma \frac{d\mathbf{r}}{dt} = -\nabla U + \sqrt{2\gamma k_B T}$$

$$-\kappa(\boldsymbol{\theta} - \mathbf{z}) \nabla \boldsymbol{\theta} \boldsymbol{\eta}^x$$

$$\bar{\gamma} \frac{dz}{dt} = \kappa(\boldsymbol{\theta} - \mathbf{z}) + \sqrt{2\bar{\gamma} k_B T} \boldsymbol{\eta}^z$$

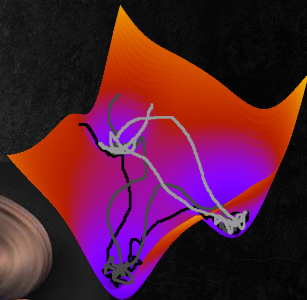
Dinámica Molecular



"Rush", Tomasz 'Darkshape' Kaluzny



"nude"
Shinichi Maruyama



Densidad de Probabilidad en el Espacio de las Fases

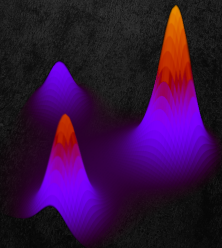
$$\rho(\mathbf{q}, \mathbf{p}, t)$$

Newton -> Ecuación de Liouville

En equilibrio $\rho(\mathbf{q}, \mathbf{p}) = \Omega^{-1}$

Langevin -> Ecuación de Fokker Planck
(integrales estocásticas)

En equilibrio $\rho(\mathbf{q}, \mathbf{p}) = \mathbb{Z}^{-1} e^{\beta H(\mathbf{q}, \mathbf{p})}$



Distribución de equilibrio

¿Qué probabilidad de encontrar un estado del sistema generan?

Newton

$$\rho(\mathbf{r}) = \Omega^{-1}, \quad NVE$$

Langevin $\rho(\mathbf{r}) = Z^{-1}e^{\beta U(\mathbf{r})}, \quad NVT$

Browniano $\rho(\mathbf{r}) = Z^{-1}e^{\beta U(\mathbf{r})}, \quad NVT$

Barostato de Andersen

$$\rho(\mathbf{r}) = Z^{-1}e^{\beta U(\mathbf{r}) + \beta PV}$$

$$NPH \rightarrow NPT$$

DM acelerada por temperatura

$$\rho(\mathbf{r}) = Z^{-1}e^{\beta(U(\mathbf{r}) + \kappa(\boldsymbol{\theta} - \mathbf{z})^2/2)}$$

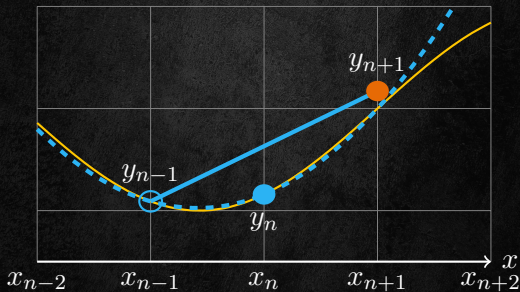
$$\rho(\mathbf{z}) \approx \bar{Z}^{-1}e^{\bar{\beta}F(\mathbf{z})}$$

¡Pero la dinámica también muestrea fuera del equilibrio!

Verlet

$$\frac{d^2y}{dx^2} = G(x, y)$$

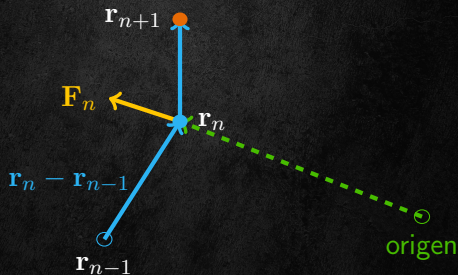
$$y_{n+1} = 2y_n - y_{n-1} + G(x_n, y_n)h^2 + O(h^4)$$



Verlet

$$m \frac{d^2 \mathbf{r}}{dt^2} = \mathbf{F} = -\nabla U(\mathbf{r});$$

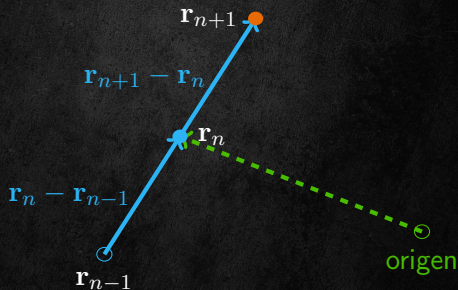
$$\mathbf{r}_{n+1} = 2\mathbf{r}_n - \mathbf{r}_{n-1} + \frac{h^2}{m} \mathbf{F}_n + O(h^4)$$



Verlet

$$m \frac{d^2 \mathbf{r}}{dt^2} = \mathbf{F} = -\nabla U(\mathbf{r});$$

$$\mathbf{r}_{n+1} = 2\mathbf{r}_n - \mathbf{r}_{n-1} + \frac{h^2}{m} \mathbf{F}_n + O(h^4)$$



Verlet

$$m \frac{d^2 \mathbf{r}}{dt^2} = \mathbf{F} = -\nabla U(\mathbf{r});$$

$$\mathbf{r}_{n+1} = 2\mathbf{r}_n - \mathbf{r}_{n-1} + \frac{h^2}{m} \mathbf{F}_n + O(h^4)$$

Velocity Verlet:

$$\mathbf{r}_{n+1} = \mathbf{r}_n + \mathbf{v}_n \Delta t + \frac{\mathbf{F}_n}{2m} \Delta t^2 + O(h^3)$$

$$\mathbf{v}_{n+1} = \mathbf{v}_n + \frac{\mathbf{F}_n + \mathbf{F}_{n+1}}{2m} \Delta t + O(h^3)$$

Movimiento Browniano y/o Ecuación de Langevin

$$\frac{dx(t)}{dt} = \sqrt{2D} \eta(t) \quad \langle \eta(t) \rangle = 0, \quad \langle \eta(t) \eta(t') \rangle = \delta(t - t')$$

Browniano Convencional

$$\mathbf{r}_{n+1} = \mathbf{r}_n + \frac{\Delta t}{m\gamma} F_n + \sigma_X \mathbf{X}_G \quad \sigma_X = \sqrt{\frac{2k_B T \Delta t}{m\gamma}}$$

$$m \frac{d^2 x(t)}{dt^2} = -\gamma \frac{dx(t)}{dt} + F(x, t) + \xi(t)$$

Ermak

$$\mathbf{r}_{n+1} = \mathbf{r}_n + c_1 \Delta t \mathbf{v}_n + c_2 \Delta t^2 \frac{F_n}{m} + \sigma_X \mathbf{X}_G$$

$$\mathbf{v}_{n+1} = c_0 \mathbf{v}_n + c_1 \Delta t \frac{F_n}{m} + \frac{c_2 \Delta t}{m} (F_{n+1} - F_n) + \sigma_V (\sigma_{V,X} \mathbf{X}_G + \sigma_{V,V} \mathbf{V}_G)$$

Integración de los momentos (NVT)

Distribución de Maxwell-Boltzmann / Espacio de configuraciones

$$\begin{aligned}\rho(\mathbf{p}, \mathbf{r}) &= Z^{-1} e^{-\beta H(\mathbf{p}, \mathbf{r})} \\&= \frac{e^{-\beta \mathbf{p}^2/2m}}{\int_{\mathbf{p}} e^{-\beta \mathbf{p}^2/2m} d\mathbf{p}} \frac{e^{-\beta U(\mathbf{r})}}{\int_{\mathbf{r}} e^{-\beta U(\mathbf{r})} d\mathbf{r}} \\&= \left(\frac{\beta}{2m\pi} \right)^{\frac{3}{2}} e^{-\beta \mathbf{p}^2/2m} \frac{e^{-\beta U(\mathbf{r})}}{Q} \\&= \tilde{f}(\mathbf{p}) q(\mathbf{r}) \quad \tilde{f}(\mathbf{p}) = f_1(p_x) f_1(p_y) f_1(p_z) \\f(v) &= \left(\frac{m\beta}{2\pi} \right)^{\frac{3}{2}} 4\pi v^2 e^{-\beta m v^2/2} \quad \langle A \rangle = \int A(\mathbf{r}) q(\mathbf{r}) d\mathbf{r}\end{aligned}$$

Otros Termostatos

Berendsen:

$$\mathbf{v}_{n+1} = \mathbf{v}_n \sqrt{1 + \frac{\Delta t}{\tau} \left(\frac{T_0}{T} - 1 \right)}$$

Andersen:

$$\mathbf{v}_{n+1} = \sqrt{\frac{k_B T_0}{m_i}} \mathbf{X}_G \quad \text{con probabilidad } \nu \Delta t$$

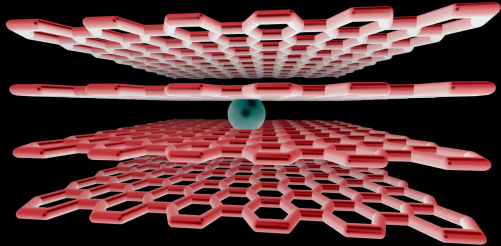
Nosé-Hoover:

$$\frac{d\mathbf{r}_i}{dt} = \frac{\mathbf{p}_i}{m_i}$$

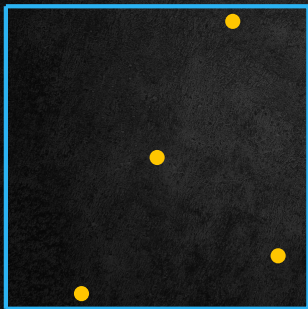
$$\frac{d\mathbf{p}_i}{dt} = -\frac{\partial U}{\partial \mathbf{r}_i} - \zeta \mathbf{p}_i$$

$$\frac{d\zeta}{dt} = \frac{1}{Q} \left(\sum_i \frac{\mathbf{p}_i^2}{m_i} - 3Nk_B T_0 \right)$$

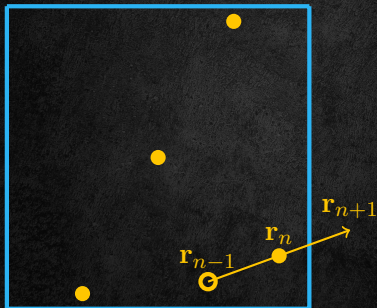
Condiciones de Contorno Periódicas



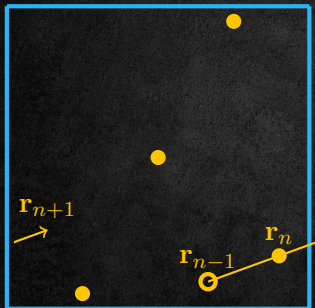
Condiciones de Contorno Periódicas



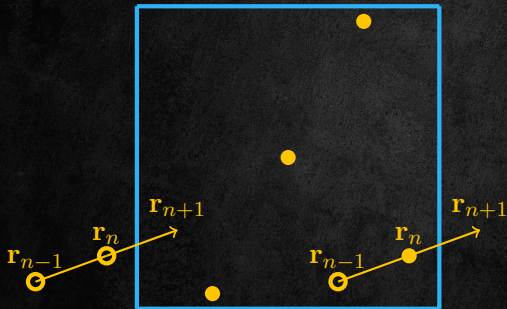
Condiciones de Contorno Periódicas



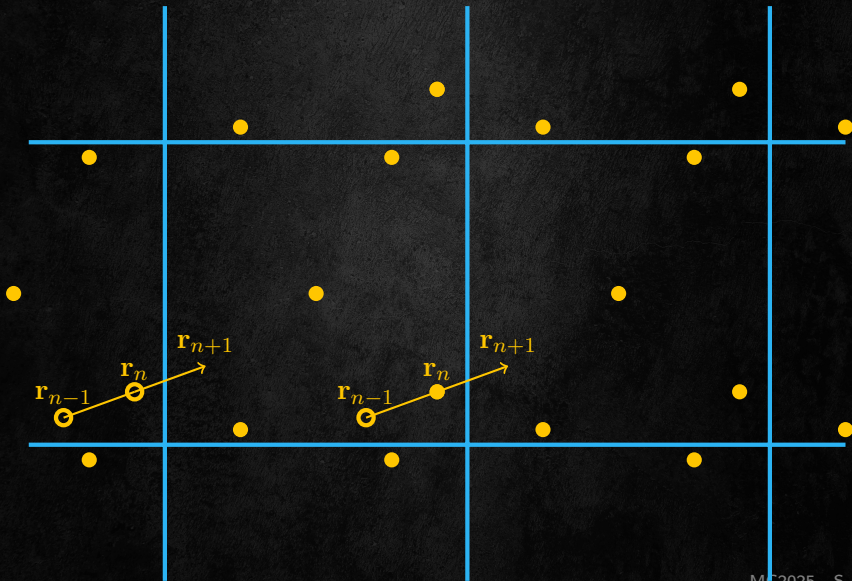
Condiciones de Contorno Periódicas



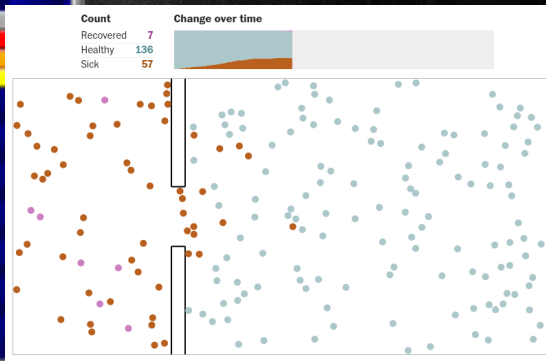
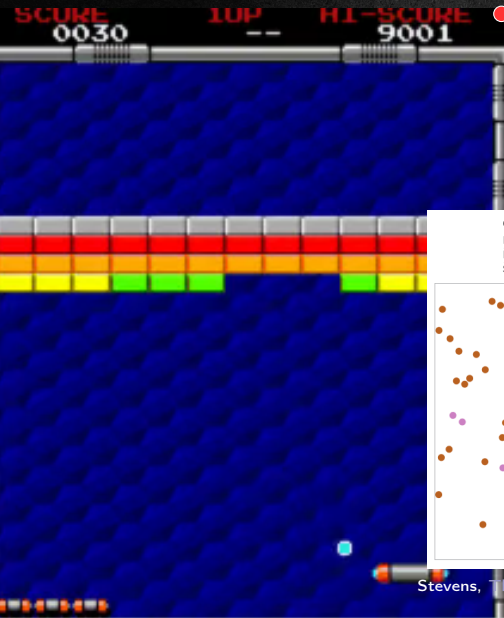
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Condiciones de Contorno Periódicas



Condiciones de Contorno Paredes Duras



Stevens, The Washington Post, March 14, 2020

Condiciones de Contorno Reservorios

